

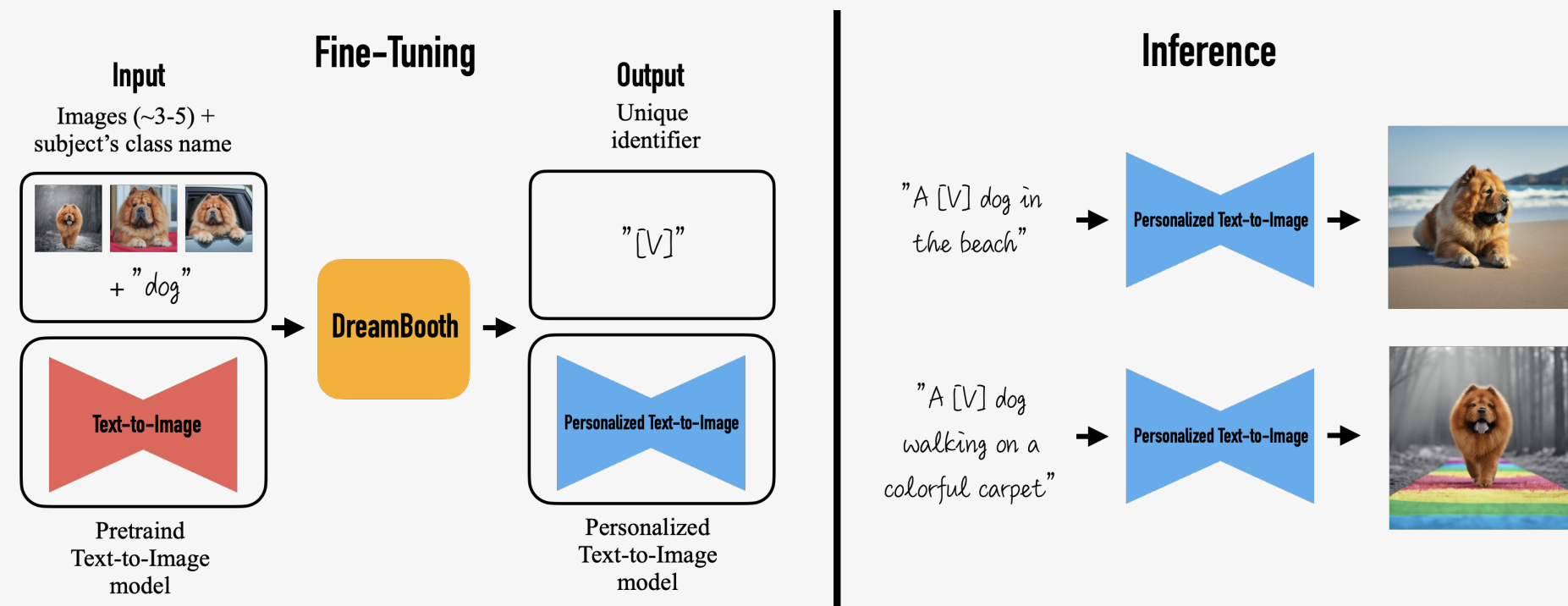
Replicating DreamBooth: Subject-Driven Fine-Tuning of Stable Diffusion

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Motivation

Generative text-to-image models produce photorealistic images from text prompts, but cannot render a *specific* subject.

DreamBooth solves this: given 3–5 photos of the subject, it fine-tunes a diffusion model, allowing it to generate images of the subject in various contexts based on the textual prompt.

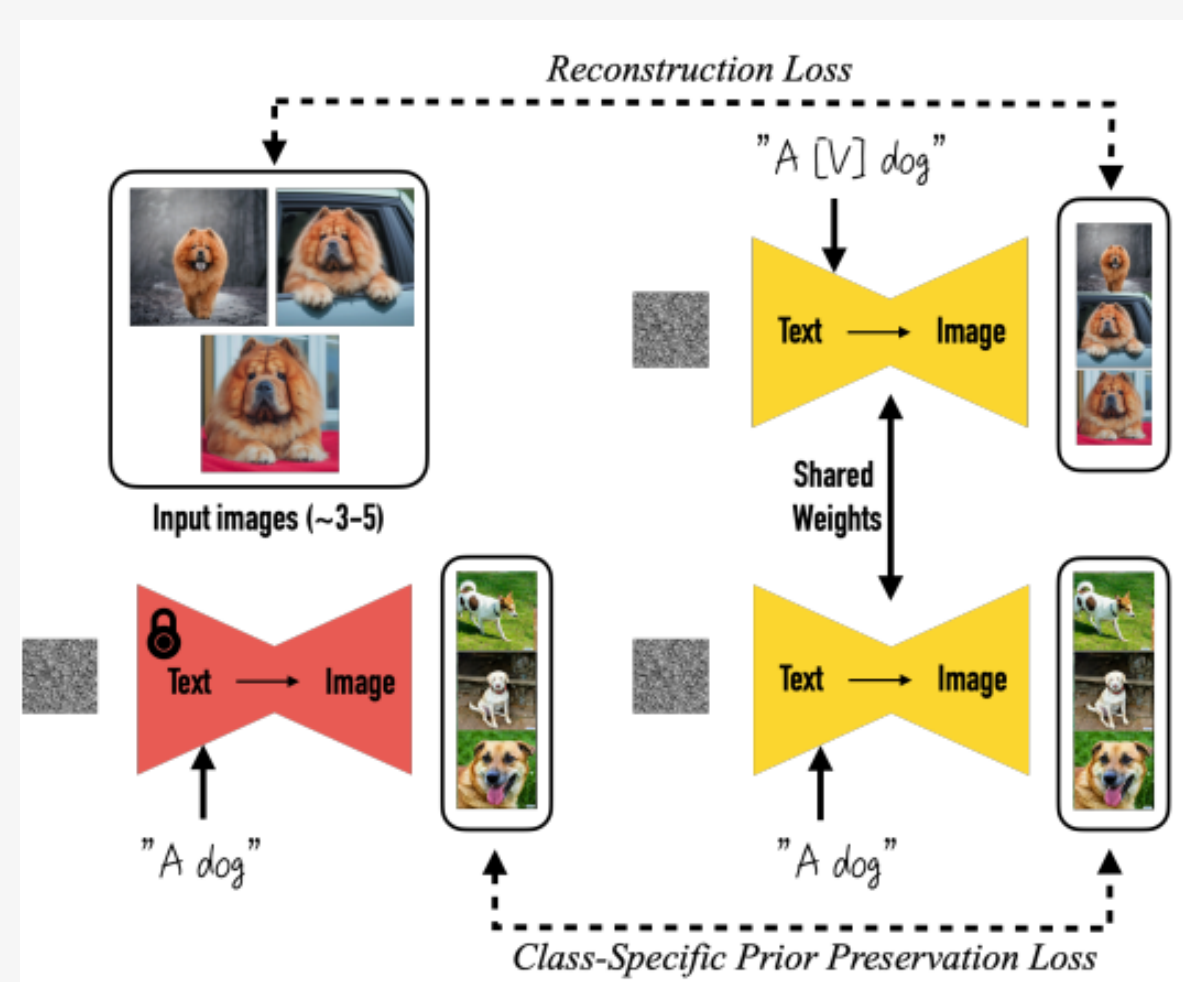


Dreambooth Technique During Fine-Tuning and Inference

Goal. Replicate DreamBooth on stable-diffusion-v1-5 (SD 1.5) and verify quantitative metrics.

Background and Key Ideas

- ▶ **Image labeling:** Label subjects with “a [identifier] [class noun]”, using **rare-token identifiers** with negligible prior meaning.
- ▶ **Prior-preservation loss:** Compare outputs with pre-generated class images to retain general class knowledge, preventing *language drift*.
- ▶ **Metrics:** Ensure *subject fidelity* with DINO & CLIP-I and *prompt fidelity* with CLIP-T



DreamBooth Loss Components and Fine-Tuning Process

Implementation Details

- ▶ **Dataset:** Substituted Google Imagen with SD 1.5 for generated samples.
- ▶ **Subjects:** dog, dog2, backpack from the official DreamBooth dataset (google/dreambooth).
- ▶ **Compute:** Google Colab T4 GPU (~30–40 min per subject). Metric computation on local CPU.

Methodology

Architecture. SD 1.5 has three components:

- ▶ **VAE** (~83M) — encodes/decodes images; *frozen*.
- ▶ **UNet** (~860M) — latent denoiser; **fine-tuned**.
- ▶ **CLIP encoder** (~123M) — encodes prompts; *frozen*.

Training loop (per step):

1. Encode instance + class images to latents via frozen VAE.
2. Encode prompts via frozen CLIP text encoder.
3. Sample noise $\varepsilon \sim \mathcal{N}(0, I)$ and timestep t .
4. Add noise: $z_t = \sqrt{\alpha_t} z + \sqrt{1 - \alpha_t} \varepsilon$.
5. Predict: $\hat{\varepsilon} = \text{UNet}(z_t, t, \tau_\theta(P))$.
6. Compute combined loss; backprop through UNet only.

$$\mathcal{L} = \underbrace{\|\hat{\varepsilon}_{\text{inst}} - \varepsilon_{\text{inst}}\|^2}_{\text{instance loss}} + \lambda \underbrace{\|\hat{\varepsilon}_{\text{class}} - \varepsilon_{\text{class}}\|^2}_{\text{prior preservation}}$$

Hyperparameters: 800 training steps, learning rate: 5×10^{-6} , resolution: 512×512 , prior weight $\lambda = 1.0$, 100 class images, 5 instance images, optimizer: 8-bit AdamW.

Additional Exploration: Logo Fine-Tuning

We fine-tuned on a custom **CS4782** course logo (5 augmented variants).

- ▶ Direct generation faithfully reproduced the logo’s structure and text with only minor color and number drift.
- ▶ Contextualization failed: placement prompts (shirt, mug) caused the model to render the raw token sks as literal glyphs rather than the learned logo.



Next steps: Augmenting with contextual mockups (e.g. logo composited onto a shirt as a reference image, paired with “a sks logo on a shirt”) might strengthen generalization.

Future Work

- ▶ **Scale to 30 subjects:** Enables direct baseline comparison.
- ▶ **Explore Hyperparameters:** Vary the learning rate and step count to identify the boundary between underfitting and overfitting.
- ▶ **Multi-subject:** Bind multiple identifiers in one fine-tune.

Results

Quantitative metrics (5 prompts $\times$ 4 samples per subject):

Subject	DINO $\uparrow$	CLIP-I $\uparrow$	CLIP-T $\uparrow$
dog	0.716	0.867	0.275
dog2	0.542	0.828	0.247
backpack	0.536	0.887	0.274
Mean	0.598	0.861	0.265
<i>Real Images</i>	<i>0.774</i>	<i>0.885</i>	<i>N/A</i>
<i>Dreambooth Paper (SD)</i>	<i>0.668</i>	<i>0.803</i>	<i>0.305</i>

Examples:



Reference subject

“a photo of [v] dog with Cornell merch”

Qualitative observations:

- ▶ Subject identity preserved across novel backgrounds and lighting.
- ▶ Fine tuning both the UNet and the text encoder leads to sub-optimal, overfit results

Conclusion

We replicated the core DreamBooth algorithm on SD 1.5, writing the training loop from scratch and verifying each component independently.

Key takeaways:

- ▶ Full UNet fine-tuning achieves strong fidelity with only 5 instance images and 800 steps.
- ▶ Metrics fall within the expected SD 1.5 range, confirming faithful replication.
- ▶ Logo fine-tuning generalizes poorly to context: direct generation reproduces the graphic faithfully, but placement prompts cause the model to render literal glyphs rather than the learned logo.

References

- [1] Ruiz et al. *DreamBooth*. CVPR 2023.
- [2] Rombach et al. *Latent Diffusion Models*. CVPR 2022.
- [3] Radford et al. *CLIP*. ICML 2021.
- [4] Caron et al. *DINO*. ICCV 2021.
- [5] <https://github.com/huggingface/diffusers>
- [6] <https://github.com/google/dreambooth>